

# Surface #1, Installment A: A Formalized Truncation-to-Envelope Bridge for the SU(3) Peter–Weyl Heat-Kernel Series

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## Abstract

We formalize, in Lean 4 against `mathlib` v4.12.0, two infrastructure lemmas needed by any cluster-expansion attack on the SU(3) lattice Yang–Mills mass gap: **(i)** summability of the Peter–Weyl heat-kernel series  $\sum_{m,n} \dim(m,n)^2 e^{-t C_2(m,n)}$  for every  $t > 0$ , and **(ii)** the bound  $S_N(t) \leq K(t)$  of every finite truncation by the genuine `tsum` envelope. Both lemmas close with axiom footprint `{propext, Classical.choice, Quot.sound}` — `mathlib`’s classical trio, no research-grade axioms. This is *not* a proof of the Clay Yang–Mills Existence and Mass Gap problem; the small- $t$  heat-kernel asymptotic that would close Surface #1 is an open analytic gap (§??), and the broader YM tower in the underlying project remains *Status: Open*.

## 1 Introduction and scope

The Kotecky–Preiss criterion for a convergent cluster expansion of the Wilson-action lattice gauge measure ultimately requires an exponential decay estimate on the per-plaquette heat-kernel activity. A standard route splits this into three pieces:

- (S) *Summability*. The Peter–Weyl series defining the heat kernel at the identity converges for every  $t > 0$ .
- (B) *Bridge*. Every finite Weyl truncation  $S_N(t)$  is bounded by the infinite series  $K(t)$ .
- (V) *Varadhan-type small- $t$  asymptotic*.  $K(t) \leq C t^{-4} e^{-c/t}$  for  $t \in (0, t_0]$  and explicit  $C, c, t_0 > 0$ .

This installment establishes (S) and (B) in Lean 4. (V) is left as an explicit open gap; see §??.

## 2 Definitions (as formalized)

The formalized definitions follow the un-normalized convention used in `Towers/YM/ClusterExpansion.lean` and `Towers/YM/PeterWeylHeat.lean`; quoted Lean names are the actual identifiers.

**Definition 2.1** (Weyl dimension for SU(3), `Weyl_dim_SU3_explicit`). For  $(m, n) \in \mathbb{N}^2$ ,

$$\dim(m, n) := \frac{(m+1)(n+1)(m+n+2)}{2}.$$

**Definition 2.2** (Quadratic Casimir, `Casimir_SU3_explicit`).

$$C_2(m, n) := m^2 + n^2 + mn + 3m + 3n.$$

(No  $\frac{1}{3}$  normalization; this is the un-rescaled lattice convention used downstream.)

**Definition 2.3** (Truncated Weyl sum, `Weyl_sum_explicit_SU3_real`). For  $t > 0$ ,  $N \in \mathbb{N}$ ,

$$S_N(t) := \sum_{m+n \leq N} \dim(m, n)^2 e^{-t C_2(m, n)}.$$

**Definition 2.4** (Heat-kernel envelope, `Heat_kernel_envelope_real`).

$$K(t) := \sum_{(m, n) \in \mathbb{N}^2} \dim(m, n)^2 e^{-t C_2(m, n)}.$$

The definition is via `mathlib`'s `tsum`; nonnegativity of every summand makes the value unambiguous on  $(0, \infty)$ .

### 3 Formalized results

**Theorem 3.1** (Peter–Weyl summability, `PeterWeyl_Summable_SU3`). For every  $t > 0$  the family  $(\dim(m, n)^2 e^{-t C_2(m, n)})_{(m, n)}$  is *Summable*.

*Proof sketch (matches the Lean argument).* The shipped bounds in `Towers/YM/PeterWeyl.lean` are intentionally slack: a linear Casimir lower bound  $m + n \leq C_2(m, n)$  and a Weyl-dim upper bound  $\dim(m, n) \leq (m + 1)^2(n + 1)^2$ . Combined with the 1D  $(n + 1)^4 e^{-\beta n}$  summability lemma applied factor-by-factor, the product-envelope squeeze gives summability for every  $\beta > 0$ . A tighter quadratic Casimir bound  $C_2 \geq c(m + n)^2$  and a cubic dim bound are flagged tripwires (project task #157); they would shorten this proof but are not needed for summability.  $\square$

**Theorem 3.2** (Truncation  $\leq$  envelope, `Weyl_sum_explicit_SU3_real_le_Heat_kernel_envelope_real`). For every  $t > 0$  and  $N \in \mathbb{N}$ ,

$$S_N(t) \leq K(t).$$

*Proof.* Every summand is nonneg, so the result is `mathlib`'s top-level `sum_le_tsum` applied to the indexing finset  $\{(m, n) : m + n \leq N\}$ , with the summability hypothesis discharged by Theorem ??  $\square$

**Theorem 3.3** (Envelope is at least one, `Heat_kernel_envelope_real_ge_one_of_pos`). For every  $t > 0$ ,  $K(t) \geq 1$ .

*Proof.* The trivial representation  $(m, n) = (0, 0)$  contributes  $\dim(0, 0)^2 \cdot e^{-t \cdot 0} = 1$ . Monotonicity of `tsum` on nonneg families closes the bound.  $\square$

### 4 Honest scope and open gaps

**Open Gap** (Small- $t$  Varadhan asymptotic for  $K(t)$ ). There is no formalized small- $t$  asymptotic

$$K(t) \leq C t^{-4} e^{-c/t}, \quad t \in (0, t_0],$$

for explicit positive constants  $C, c, t_0$ . The classical statement (Molchanov / Varadhan for compact semisimple Lie groups, with exponent  $-\dim G/2 = -4$  for  $SU(3)$ ) is widely cited but no machine-checked proof has been produced in this project. It is queued as project task #156, and until it lands no quantitative plaquette-activity bound of the form  $\text{activity}(\beta) \leq C \beta^{-4} e^{-c/\beta}$  has been proved here.

**Open Gap** (Tightening of the Casimir and Weyl-dim bounds). The lower bound  $C_2(m, n) \geq c(m + n)^2$  and the cubic-in- $(m + n)$  dimension bound used in the informal literature are not in the current Lean spine; only their linear / product-square slack versions are shipped. Project task #157 covers the upgrade.

**Open Gap** (Plaquette activity bound and Kotecky–Preiss closure). The chain

$$\text{exists\_c\_per\_plaquette\_pw} \Rightarrow \text{polymer\_activity\_bound\_real\_pw} \Rightarrow \text{kotecky\_preiss\_criterion}$$

depends on the Varadhan gap above. Until that closes, Surface #1 is not complete and the Yang–Mills tower in the project's roadmap (`docs/ROADMAP.md`, §2) remains Status: Open.

## 5 What the present installment does and does not claim

**Claims:** Theorems ??, ??, and ?? are formalized in Lean 4 against `mathlib v4.12.0` in `Towers/YM/PeterWeyl.lean` and `Towers/YM/PeterWeylHeat.lean`. They are sorry-free and close with axiom footprint `{propext, Classical.choice, Quot`. A previously parked sorry at `Towers/Attempts/ClusterExpansion.lean:693` is discharged by retargeting its conclusion from a mathematically false placeholder (`Heat_kernel_def_real`, the Varadhan asymptotic shape, which fails at  $N = 0, t = 1$ :  $1 \not\leq 0.368$ ) onto the genuine `Heat_kernel_envelope_real` of Theorem ??.  
**Does not claim:** a proof of the  $SU(3)$  Yang–Mills mass gap, completion of Surface #1, the Varadhan small- $t$  asymptotic, the plaquette-activity bound, or any quantitative spectral-gap statement.

## Reproducibility

The cumulative project wall, after the two batches that landed this installment (project tasks #154 and #155), is 460 trio-clean bricks under `scripts/check-towers.sh`; the relevant subset is 8 bricks (4 in `Towers/YM/PeterWeyl.lean`, 4 in `Towers/YM/PeterWeylHeat.lean`). The Genesis-sealed append-only ledger `data/hits.txt` verifies against SHA-256 `eecbcd9a540aa7a2c90edd23827c73e4d1bb5af641d352f70a5de849b21f875f` at the time of writing; verification by `scripts/check-genesis-seal.py`.